

A decision support framework to incorporate textual data for early student dropout prediction in higher education

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ABSTRACT

Managing student dropout in higher education is critical, considering its substantial impacts on students' lives, academic institutions, and society as a whole. Using predictive modeling can be instrumental for this task, as a means to identify dropouts proactively on the basis of student characteristics and their academic performance. To enhance these predictions, textual student feedback also might be relevant; this article proposes a hybrid decision support framework that combines predictive modeling with student segmentation efforts. A real-life data set from a French higher education institution, containing information of 14,391 students and 62,545 feedback documents, confirms the superior performance of the proposed framework, in terms of the area under the curve and top decile lift, compared with various benchmarks. In contributing to decision support system research, this study (1) proposes a new framework for automatic, data-driven segmentation of students based on textual data; (2) compares multiple text representation methods and confirms that incorporating student textual feedback data improves the predictive performance of student dropout models; and (3) establishes useful insights to help decision-makers anticipate and manage student dropout behaviors.

1. Introduction

Student dropout has been one of the most important topics in education and academic research [1,2]. Along with the development of educational models, this persistent problem becomes more and more diverse. Studies on student dropout, therefore, are also divided into several categories, focusing on different aspects of the problem, such as the levels of education (e.g., primary, secondary, tertiary education, etc.), the studying environments (e.g., on site, online, mixed, etc.), and/or the scale of studying (e.g., course, semester, program, etc.) [3–7]. Dropping out also manifests at different scales, from a single course (including online course) which may need the attention of the teacher, to an entire program leaving more serious consequences. Within this study, we investigate the dropout problems of students for an entire program at tertiary level in higher education institutions (HEIs) since this is the level of education in which dropping out is common, and if it happens, leaves significant impact to learners and stakeholders [8,9]. When students drop out of their HEIs programs, they suffer resource losses, in terms of time, effort, and financial investments [4,8]. Society at large also suffers, because graduation rates tend to influence unemployment rates, income tax contributions, and social activities [10]. The

HEIs need to maintain low student dropout rates to ensure high rankings, maintain accreditations, and enhance external perceptions among prospective students [2,11]. Most institutions thus adopt initiatives to provide support for students, such as tutoring services [9]. Yet more than one-quarter of students who start a four-year curriculum do not obtain their degrees [12]. The dropout behaviors also vary depending on the peculiarities of the national education system as well as the quality of the institution's educational environment [13,14]. Institutions with high international rankings might have low dropout rates compared to others [15].

Beyond supportive initiatives, HEIs also rely on decision support frameworks and predictive modeling to anticipate student dropouts [2]. This modeling task is usually carried out by a binary classification setup that trains machine learning methods to classify new probabilistic student observations in two groups: dropout and non-dropout. This setup relies on historical student data to predict future student dropout probabilities, which decision-makers leverage to undertake a smart, data-driven selection of which students to target with retention campaigns [16]. The availability of various rich data sources offers new opportunities for such data-driven decision making. Data often are classified as structured or unstructured [17], such that structured data

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can be stored in tabular formats in a relational database with a clear metadata structure [18,19] (e.g., student demographics, needs; classroom characteristics; cognitive, academic and behavioral engagement variables). These data often are numerical [3]. Unstructured data instead do not have a metadata structure and need to be converted into a structured format before they can be modeled [18,20], such as text, images, audio or video files. They comprise approximately 80% of data stored in any organization's databases [21,22].

For example, HEIs gather substantial textual data from student reviews, feedback, or posts on the schools' online learning platforms. Such textual student feedback data offer notable promise for student dropout prediction, because in these data, students freely express their opinions of learning processes, facilities, courses, and academic staff. In addition, textual data have proven valuable for other predictive modeling applications, such as those related to customer churn [23], financial risk [24] or even student dropout [25]. High-dimensional representations of textual data capture information that is not available from structured data sources; in this sense, student feedback might improve the performance of existing student dropout prediction models by adding new information. Unstructured textual data even may be particularly well-suited for segmentation efforts to improve predictive performance [26]. Student textual feedback likely can reveal different groups of otherwise similar students, so from a predictive modeling perspective, textual data might be able to uncover latent student groups with potentially different dropout drivers. Despite that, there have not been many research efforts on incorporating unstructured textual data to student dropout predictions and segmentation in higher education contexts. In addition, previous research acknowledges that a segmented approach to student

dropout prediction can yield significant improvements in predictive performance. For example, Coussemment et al. [3] propose a logit leaf model (LLM), a hybrid algorithm based on decision trees (DT) and logistic regression (LR), in a subscription-based online learning context. The LLM algorithm automatically creates segments of students and can detect segment-specific dropout drivers, so it offers a good balance between *predictability* and *interpretability* [27]. However, previous studies tend to rely solely on structured data for segmentation efforts and neglect unstructured data. Therefore, we incorporate textual feedback data into a student dropout prediction application, using text representation and segmentation methods, in an attempt to answer the following research questions:

RQ1. Does incorporating textual feedback data improve student dropout prediction?

RQ2. What insights can be derived from the incorporated textual data to help decision-makers anticipate and manage student dropout?

To test our proposed approach, we also gather five years of student data from a HEI in France, which span 14,391 students and 62,545 textual feedback documents. In turn, we make three main contributions. First, we propose a new framework for automatic, data-driven segmentation of observations that is based on unstructured, textual data. Second, we introduce unstructured, textual feedback data to student dropout prediction application. Incorporating such data enables us to compare multiple text representation methods and assess their value, relative to structured data. We also validate the performance of a predictive model, based on the proposed framework, using real-life data

Table 1
Studies predicting student dropouts from HEI using predictive modeling, published 2009–2021.

Authors	Context	Data Set	Data Type		Modeling Approach		Interpretability	
			Structured	Textual	Algorithm	Segmentation	Global	Segment
[7]	Freshmen dropout, public university in Netherlands	648 students, 49% dropouts	✓	-	DT, BN, LR, RF, RL, OneR	-	✓	-
[32]	Student at risk, public university in UK	4223 students, n/a dropouts	✓	-	NB, SVM, DT	-	✓	-
[16]	Freshmen dropout, public university in US	16,066 students, 21.84% dropouts	✓	-	DT, NN, SVM, LR	-	✓	-
[33]	Freshmen dropout, public university in US	25,224 students, 21.03% dropouts	✓	-	DT, NN, LR	-	✓	-
[74]	Freshmen retention, public university in US	n/a students, 28.7%, 39.6%, 45.2% dropouts after 1st, 2nd, 3rd year	✓	-	OneR, DT, NB, BN, NN	-	✓	-
[30]	Student dropout, public university in Czech Republic	775 students, 58.86% dropouts	✓	-	NB, SVM, RL, OneR, DT, LL	-	-	-
[75]	Freshmen dropout, private university in US	429 students, 11.5% dropouts	✓	-	NB, DT, LR, RF	-	-	-
[1]	Freshmen dropout, public university	21,654 students, 21.3% dropouts	✓	-	NN, SVM, DT, LR	-	✓	-
[69]	Student dropout, public HEI in US, STEM program	12,293 students, n/a dropouts	✓	-	SA, LR, DT, NB, AB	-	✓	-
[28]	Freshmen dropout, public university	682 students, n/a dropouts	✓	-	LR, NN, PNN	-	✓	-
[34]	First year student attrition, public university in US	66,060 students, 28.40% dropouts	✓	-	LR, kNN, RF, SVM, GBT	-	-	-
[76]	Student at risk, public university in Canada	38,842 students 31.65% dropouts	✓	-	RF	-	✓	-
[29]	Student dropout, university in Brazil	2175 students, 13.2% dropouts	✓	-	NB, SVM, RF, GBT, kNN	-	-	-
[25]	Student dropout, university in US	7343 students, n/a dropouts	-	✓	RF, SVM, LR, DT	-	-	-
[2]	Freshmen dropout, public university	36,461 students, 20% dropouts	✓	-	BN	-	✓	-
[6]	Student dropout, public university	3176 students, 19.5% dropouts	✓	-	LR, DT	-	✓	-
[31]	Student at-risk (withdrawn/dropout), public university in Canada	25,541 students, n/a dropouts	✓	-	DL, SVM, LR	-	-	-
[9]	Student dropout, HEI in Chile	3362 students 12.4%, 5.4% dropouts after the 2nd, 3rd year	✓	-	LR, kNN, DT, RF, NN, SVM	-	✓	-
This study	First year student dropout private business school	14,391 students, 1.2% dropouts	✓	✓	LR	✓	✓	✓

Notes: The predictive algorithm abbreviations are as follows: adaptive boost (AB), Bayesian belief network (BN), deep learning (DL), decision tree and its variants (e.g. CART, C4.5, J4.8) (DT), gradient boosted trees (GBT), k-nearest neighbor (kNN), lazy learning (LL), logistic regression (LR), naïve Bayes (NB), neural network (NN), one rule (OneR), probabilistic neural networks (PNN), random forest (RF), rule-based learner (RL), support vector machines (SVM), and survival analysis (SA).

and in comparison with benchmark approaches. Third, we demonstrate that these unstructured, textual data contain actionable information that can support decision-makers, on the basis of model interpretation and visualization.

2. Literature review

We present an overview of peer-reviewed literature between 2009 and 2021 pertaining to student dropout predictions in HEIs that uses predictive modeling approaches in Table 1; it reveals three main aspects that are relevant for our study. First, the *input data* used thus far to model and predict student dropouts include mostly variables derived from structured data sources, usually the HEI's relational database. The two most frequently investigated sets of structured variables are students' background information and academic performance. The former category includes sociodemographic information, such as students' age, gender, nationality or ethnicity, parents' jobs, and whether they require financial aid [2,6,28]. This information is available at the start of students' academic careers and unlikely to change over time. Kemper et al. [6] report that older (age) and German (nationality) students tend to drop out more; Mason et al. [28] claim that financial aid acceptance and amounts (financial aid) and family income levels (parents' jobs) also increase dropout likelihood. Gender does not emerge as an important predictor in either study. Variables related to academic performance, prior to or during current literature, include school performance, course/program enrollments, grades/GPA, acquired/failed credits, failed courses, number of studying hours, number of retakes, and class ranking [1,2,29]. Delen et al. [2] find that students whose GPA is equivalent to an A have an only 7.3% probability of attrition, whereas students who receive an F average exhibit a much higher probability, of 87.8%. Manrique et al. [29] predict student dropout with high accuracy and F1 scores, using only their academic performance; Thammasiri et al. [1] find that GPA, rate of hours earned over hours registered, SAT scores, scholarships, and financial aid are among the top 10 predictors for dropout prediction. A few studies also creatively incorporate other sources of structured data, such as when Bayer et al. [30] construct a social graph based on students' discussion boards, and Waheed et al. [31] leverage clickstream data that reflect student interactions with online learning platforms to identify at-risk students. Unstructured textual data, on the other hand, have been rarely included for predicting student dropout, despite being available from many institutions. Zhang et al. [32] develop a data mining system to predict students at-risk of dropping out which relies on rich sources of educational data. Though, the collected textual data is limited to descriptions of courses and modules and is not used to predict dropout but only serves as input for a post-hoc analysis. Jayaraman [25] demonstrates that the dropout behaviors can be predicted with high accuracy by using solely the student advisors' text notes. However, this study does not investigate the possibilities of incorporating textual data with structured data as well as using it to detect students' segments.

Second, several *predictive modeling* algorithms inform student dropout predictions. Our literature review indicates that LR is the most popular algorithm and achieves robust prediction performance. According to Delen [33], LR, decision tree (DT), and neural network (NN) can predict dropouts by students in an unbalanced data set (21.03% dropout) with accuracies between 70% and 80% and recall metrics between 63% and 68%. Thammasiri et al. [1] compare four models: LR, DT, NN, and support vector machines (SVM), with and without sampling techniques. Recent studies also investigate the application of more complex machine learning methods. For example, Aulck et al. [34] show that LR performed equivalently with ensemble methods (random forest [RF] and gradient boost tree [GBT]) for student attrition prediction. Waheed et al. [31] find that deep learning (DL) outperforms traditional shallow machine learning models when it includes student clickstream data, though at a huge cost of additional complexity for the model tuning and deployment. Our literature review also indicates that most

previous work groups students into a single homogeneous cohort (*global*), instead of predicting dropout behavior for different groups (*segment*). To advance decision support system and learning analytics literature, we thus propose a decision support framework that allows for the automatic segmentation of students, on the basis of their textual feedback data, before applying a LR to each segment. We present the details of this framework in Section 3.

Third, the *interpretability* of prediction models is critical to data science in general [35] and to decision-makers who seek to manage student dropout in particular [3,6]. But it depends on human interpretation and cannot be described with a pure mathematical formula. For this study, we define interpretability as the degree to which humans can understand the decision recommended by the algorithm [36], and we distinguish *global* and *segment* interpretability. The former offers insights into the model's predictions over all observations; the latter supports interpretations at the *segment* level. A *global* interpretation can pertain to all types of prediction models, suggesting a capacity to extract and analyze the model structure and/or internal statistics directly (e.g., coefficients, decision rules), or else use a post hoc method, such as permutation feature importance or the Shapley Additive exPlanations (SHAP) method. In contrast, a *segment* interpretation should reveal different group characteristics that remain unknown at a *global* level and that depend on whether the prediction model has the ability to create segments or not [27]. As Table 1 indicates, various studies offer interpretations of their prediction models and offer insights about student dropout drivers. For example, Delen [33] reports the most important variables based on a sensitivity analysis, and Thammasiri et al. [1] cite feature importance scores for the DT model. Both Mason et al. [28] and Kemper et al. [6] explain dropout behaviors using the beta coefficients of their LR model. In more recent work, Delen et al. [2] propose a Bayesian belief network (BBN) that captures probabilistic interactions and nonlinear relationships between the dependent variable and the independent variables. These studies thus consider a *global* scope of interpretability; with our decision support framework, we seek to allow for interpretation at the *segment* level and thereby provide decision-makers with insights into the rationale for different predictions about different groups of students.

3. Methodology

In presenting our hybrid decision support framework, to incorporate textual data into predictive modeling efforts, we start in Section 3.1 by introducing the framework used to predict student dropout (Fig. 1), which leverages both structured and unstructured textual data. Section 3.2 describes the representation methods we propose to handle unstructured textual data. Then in Sections 3.3 and 3.4, we describe the methodological building blocks for our framework.

3.1. Framework to incorporate textual data

As Fig. 1 reveals, the proposed framework contains two stages: segmentation and modeling. Its input is data about the set of all students $DT = (X^S, X^U, Y)$, which includes structured data X^S , unstructured textual data X^U , and binary student dropout status Y . The unstructured textual data X^U are processed through a text pre-processing pipeline and transformed into a structured format, according to a text representation method. In the first, segmentation stage, we create homogeneous segments of students on the basis of the transformed textual data. That is, we specifically use unstructured, textual data to construct student segments and maximize interpretability, while avoiding issues associated with using mixed data for distance-based segmentation [37]. The transformed textual data serve as input for a segmentation algorithm to create K homogeneous segments; on the basis of these created segments, we can divide the processed, structured data and transformed textual data into K subsets, $DT_{k \in [1..K]} = (X_k^S, X_k^U, Y_k)$.

In the second, classification stage, we train models for each segment

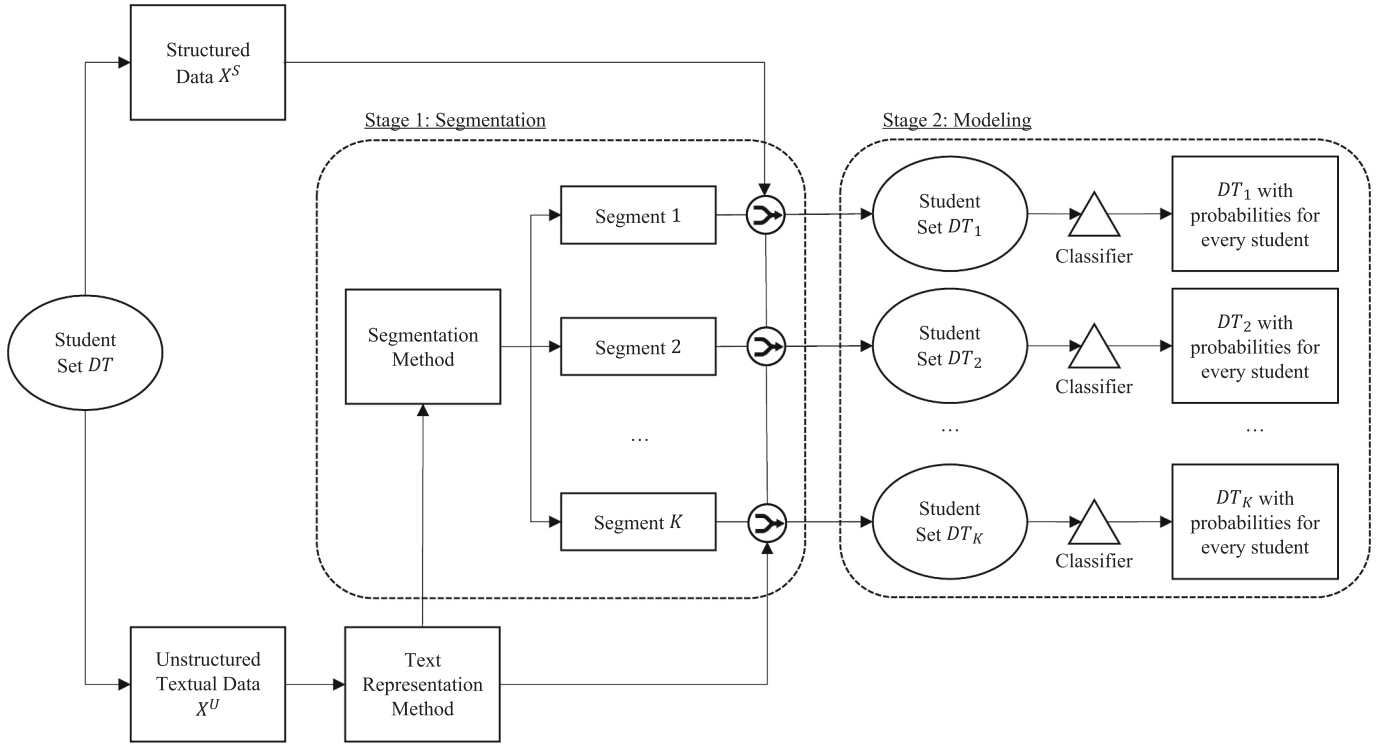


Fig. 1. Decision support framework to incorporate textual data to predict student dropout.

separately. For each subset $DT_{k \in [1..K]}$, the structured data X_k^S and transformed textual data X_k^U get combined and serve to train a predictive model that can produce segment-specific dropout probabilities. The output of this stage is a unique predictive model for every student segment.

The selection of methods for the two stages of the proposed decision support framework can be flexible, but for this study, we use vector space, doc2vec and BERT [38–40], for text representation, and k-means and hierarchical clustering algorithms for clustering. Then in the modeling stage, we prefer LR, due to its good prior performance and interpretability. We provide more information about these selected methods in Sections 3.2–3.4.

3.2. Text representation methods

We adopt three methods to represent textual data, the vector space approach, the doc2vec approach and the state-of-the-art transformer language model BERT [38–40]. The transformed textual data serve as input to create student segments but also as features to train the dropout prediction model (Fig. 1).

3.2.1. Vector space

The vector space approach provides a baseline method, reflecting its popularity and strong performance [20,23]. It features two main steps: term-vector weighting and dimension reduction [39,41]. The term-vector weighting method represents each textual feedback document as a numerical vector in a feature space, using weighted term frequencies [39]. This method can represent a collection of feedback using a list of unique terms, as well as identify the most informative terms on the basis of their weights. The weighting vector of each term t_i in the feedback document d_j can be determined by calculating the term frequency (TF_{ij}) and the inverse document frequency (IDF_i), in which $i \in [1..M]$ where M is the number of unique terms in the feedback document corpus, and $j \in [1..N]$ where N is the number of feedback documents. The TF values measure the importance of terms in representing the content of a feedback document, while the IDF adjusts the impact of

common (rare) terms in the feedback document corpus. The term weight v_{ij} of the term t_i in the feedback document d_j , therefore, is calculated by multiplying the corresponding TF_{ij} with the IDF_i values, as in Eq. (1):

$$v_{ij} = TF_{ij} * IDF_i \quad (1)$$

Thus, a feedback document d_j can be represented by a vector of corresponding term weights $\vec{v}_j = (v_{1j}, v_{2j}, \dots, v_{Mj})$ in the feature space, and the collection of all feedback vectors is the $TF - IDF_{M \times N}$ term-feedback document matrix.

In a second step, the term-feedback document matrix, usually high dimensional but sparse, is transformed into a low-dimensional space based on latent semantic indexing (LSI), using a singular value decomposition (SVD) method [41]. Using LSI, based on SVD dimensionality reduction, not only helps reduce noise but also improves predictive performance [42,43]. In detail, LSI anticipates l latent semantic concepts in the corpus. The term-feedback document matrix gets projected from a high dimensional space to a lower dimensional, semantic latent subspace, because we group the terms into l distinct concepts. This step relies on a truncated SVD that can decompose the weighted term-document matrix into three distinct matrices:

$$TF - IDF_{M \times N} \approx U_{M \times l} \cdot \Sigma_{l \times l} \cdot (V_{N \times l})^T, \quad (2)$$

where $U_{M \times l}$ and $V_{N \times l}$ are orthogonal matrices, and $\Sigma_{l \times l}$ is a non-negative diagonal matrix. Furthermore, $U_{M \times l}$ and $V_{N \times l}$ represent the relationships of the M -terms and N -feedback documents with the l latent concepts, respectively, and the diagonal of $\Sigma_{l \times l}$ indicates the importance of the latent semantic concepts in the l dimensional subspace. Finally, to find the optimal number of l semantic concepts, we use the profile log-likelihood method proposed by Zhu and Ghodsi [44].

3.2.2. doc2vec

The document representation method doc2vec derives directly from the word representation method word2vec [45,46]. word2vec method is a self-supervised framework mapping every word in the vocabulary to a unique vector in the representation vector space so that similar words

are located close to each other. This representation model can be trained by two ways, in which, each word either is predicted by the surrounding words (CBOW method) or used to predict other words in the same context (Skip-gram method). By using neural networks with new architectures, word2vec can preserve the linear regularities among words, and is able to capture syntactic and semantic word similarities better than other methods [45]. As a natural extension, doc2vec vectorization, introduced by Le and Mikolov [38], represents a feedback document corpus according to a continuous distributed vector. Previous studies show that it is more robust and accurate in document classification than other baseline methods [38,47,48]. Similar to word2vec, there are two main architectures to train the doc2vec model: a distributed memory model of paragraph vectors (PV-DM) and the distributed bag of words version of the paragraph vector (PV-DBOW).

3.2.3. BERT

Bidirectional Encoder Representations from Transformers (BERT) is a transformer-based language model introduced by Devlin et al. [40] in 2018 as a state-of-the-art method in natural language processing research. Unlike vector space and doc2vec, BERT applies a masked language model that enables bidirectional training from text and allows the model to capture deeper sense of the context [40,49]. Like modern language models, a pre-trained model of BERT is available that can be finetuned for different tasks. In this study, we select and compare three common variants of BERT from the TensorFlow Hub, which are the BERT Base (transformer blocks $L = 12$, hidden size of $H = 768$, attention heads $A = 12$), the Small BERT ($L = 4$, $H = 512$, $A = 8$), and the ALBERT ($L = 12$, $H = 768$, $A = 12$).

3.3. Segmentation methods

In this study, we test two common unsupervised methods for segmentation, i.e., k-means and hierarchical clustering. On the one hand, k-means segments the input data into K different clusters [50]. These clusters are randomly initialized, then updated in several iterations to minimize the within-cluster variation, as calculated according to the Euclidean distance. On the other hand, hierarchical clustering uses a bottom-up approach in our study and initiates every single input data point as singleton cluster. The algorithm then sequentially merges pairs of clusters that are closest together. Euclidean distance is used to measure the distance between clusters' centroids [51]. A challenge to using the k-means and hierarchical clustering algorithm is the need to select an optimal and/or reasonable number of clusters (K). However, since segmentation is an integral step in the decision support framework, the optimal number of K segments depends on the predictive performance of the framework, according to a cross-validation setup. We apply segmentation methods only to transformed textual data with the same value range, so it is not necessary to scale the input [50].

Furthermore, the segmented and transformed textual data can be visualized and evaluated using various visualization tools. In this study, we use three most common tools to present the results i.e., principal component analysis (PCA), t-distributed stochastic neighbor embedding (t-SNE) and uniform manifold approximation and projection (UMAP). These visualization tools, although different in methodology, have the same goal to reduce the complexity of the high-dimensional representations of textual data while preserving the most important information to validate the detected segments. PCA is the oldest yet most common dimensional reduction method that linearly transforms the original variables to a new set orthogonal variables (a.k.a. principal components) where the data variance is compressed and stored principally in the first few components. Plotting these first two or three components gives an overview of what the data look like [52,53]. t-SNE is a visualization technique based on stochastic neighbor embedding (SNE) that maps data points, which are described by pairwise dissimilarities, from high-dimensional space to low-dimensional space in a way that preserves local structure. For that reason, t-SNE plots tend to show data clusters

thus is very useful in segmentation-oriented applications [54,55]. Lastly, we use the recent dimension reduction and visualization tool: UMAP. This state-of-the-art method constructs two fuzzy topological representations of the data in high and low dimensional spaces, then optimize the data representation in low dimensional space to minimize the cross-entropy between two of them [56,57].

3.4. Logistic regression

Logistic regression is one of the most popular classification method in learning analytics [58], and especially for predicting student dropout [59–62]. This popularity arises because (1) the output of LR is interpretable, in that it reveals the linear relationship of a variable with the log-odds of the dependent variable; (2) it provides good and stable prediction results; and (3) it is efficient to train. Furthermore, LR can handle large amounts of structured and unstructured variables with its regularization method, such as the shrinkage Lasso (L1) [63]. This L1 method also acts as a variable selection step, which helps reduce the complexity and increase the stability of the LR model.

4. Research procedure

In this section, we detail the research procedure, starting with a description of the real-world data set and a definition of student dropout. After we present the data processing pipeline for structured and unstructured textual data, we describe our experimental setup: timeline for the independent and dependent variables, the 5×3 cross-validation setup, the hyperparameter tuning method, selection process, and data segmentation strategy. Finally, we outline the evaluation metrics.

4.1. Data

This study uses real-life data of 14,391 students, provided by a French HEI. The data cover five consecutive academic years, from 2013 to 2018, and refer to both bachelor's and master's degree program students. The structured data refer to students' sociodemographic information, program and course enrollment information, grades, and study status. The unstructured data include their textual feedback about each of their enrolled courses. Students provide this textual feedback at several moments in the academic year, as part of course evaluations. About half of the sample (7226 students, 50.2%) provided detailed textual feedback, and each student evaluates an average of eight courses.

Student dropout after the first year is a major issue; the first semester is the most intense period for most students [2,64,65]. We identify an early dropout as a student who completely quits school during the second or third semesters of the program. Fig. 2 depicts the time window of the experiment, in which the independent period is always the student's first semester (S1), which starts in the last week of August and ends just before the Christmas holidays in December; the dependent period covers the two next semesters (S2, S3), starting in the following January and lasting until the end of the third semester (next December). The holiday break, between the first and second semester, is a gap period. By using available information in their first semester of the first year at the HEI, we attempt to predict if students will drop out in the second or third semester; during the five-year period, 1.2% of all students dropped out.

4.2. Data processing

The data processing involves three steps (Fig. 3). First, the raw structured data are extracted from the institution's database to create a student basetable. These raw data contain information about students' studying activities, academic performance, and sociodemographics. Aggregated to the student level, they represent the student's first semester of the first year. Missing values are assigned a value of 0 or mean imputed, depending on the variable. Outliers, or observations that differ

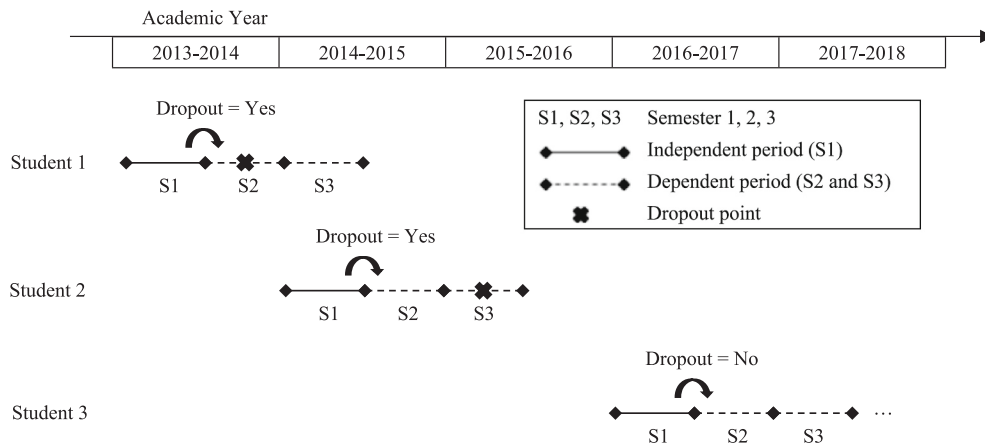


Fig. 2. Time windows for early student dropout prediction.

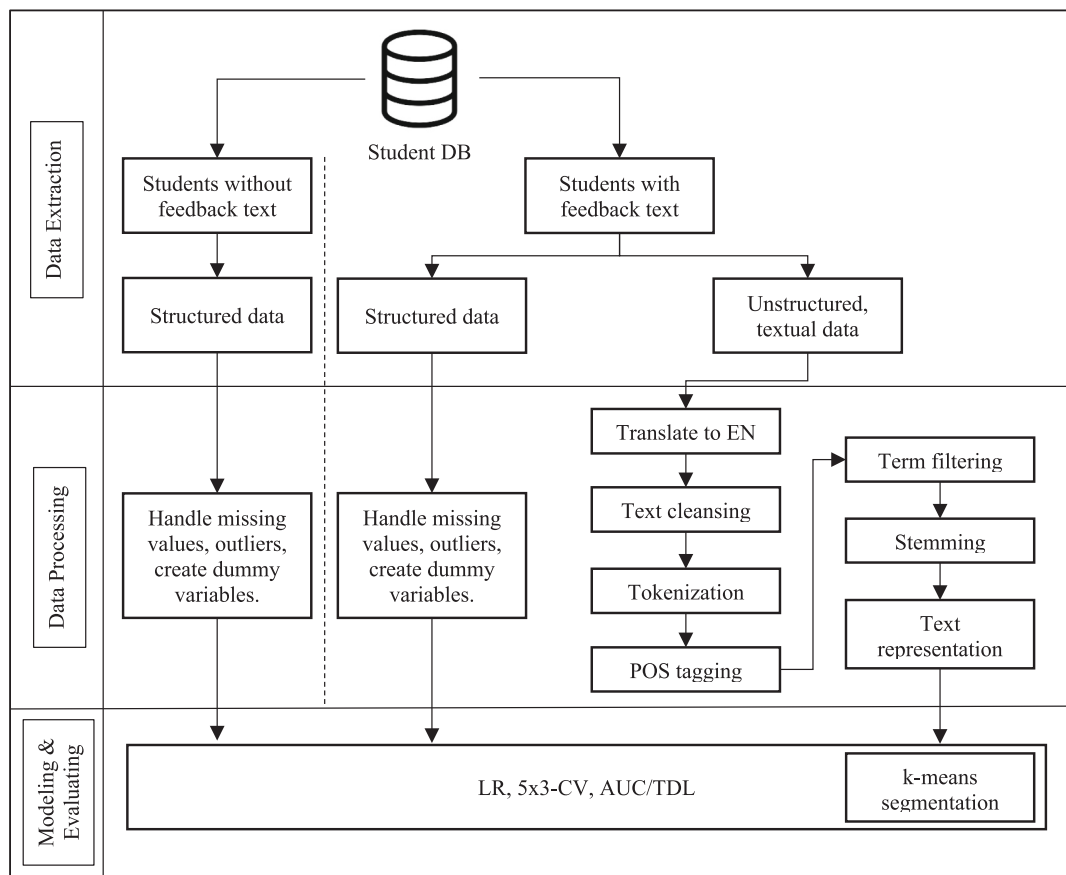


Fig. 3. Data processing pipeline.

significantly from others, are imputed for each variable in the student basetable, using the mean $\pm$ 3 standard deviations. Finally, any categorical variables get converted to dummy variables.

Second, the raw, unstructured, textual data come from student evaluation forms, completed after every course. We gather the feedback text at the student level, to derive a single document per student. These raw texts are processed, step by step, as in Fig. 3. To clean the text, we removed special characters and punctuation, then converted the cleaned text to lowercase and tokenized it on the word level. Part-of-speech tagging (POS), using the Penn Treebank tag set, identifies the grammatical combination, filtered according to informative tokens (i.e., nouns, verbs, adjectives, and adverbs) [66,67]. In this step, stop words

and low frequency words, which appear <3 times, are dropped. Finally, the cleaned words are stemmed according to a Snowball method [68].

Third, the tokenized, unstructured, textual feedback data need to be transformed into document vectors. Different methods are available for this step, so we compared the most popular text representation method, being vector space, doc2vec and BERT. The number of dimensions of the embedding vector is based on the cross-validation. Table 2 presents an overview of all variables in the data set.

4.3. Experimental setup

The processed, structured data and the transformed, unstructured,

Table 2
List of variables.

No.	Variable Group	Variable Description	# Variables
1	Sociodemographic information	Gender (Male/Female)	1
2		Citizen status	1
3		Civil status	1
4		Continent	1
5		2 first digits of zip code	1
6		Age	1
7		Information change (Yes/No)	1
8		Campus (Lille/Paris)	1
9		Number of courses enrolling in this semester	1
10	Course enrollment	Number of courses enrolled by category (Accounting, Economy, Finance, Marketing, Negotiation, Language, etc.)	23
11		Number of courses retaking in this semester	1
12		Number of internships having in this semester	1
13		vector space matrix doc2vec matrix BERT matrix	Decided by hyperparameter tuning
14	Target variable	Student dropout after the first semester (Yes/No)	1

textual data then combine to provide the inputs for the experimental setup. Table 3 contains all these setups, listing the groups, indexes, and names of the models; the data from students with (T) or without (NT) textual data; and the two types of data. The unstructured, textual data type, denoted U in Table 3, is only available for the student group with textual data, by definition. The table also reveals the representation method, vector space, doc2vec or BERT method, and k-means or hierarchical clustering segmentation method. By combining those elements, we create 7 models, which we label based on their elements. In turn, we can classify the models into three groups, which we use to assess the implications of incorporating textual data for student dropout predictions. The models in Group 1 reflect the differences between the two student groups, T and NT, which would imply benefits to modeling them separately. Groups 2 and 3 focus on students that provide feedback text T to gauge the value of U data type and the improvement in student dropout prediction if we incorporate this type of data and its segmentation layer.

The models are trained and tuned using 5×3 cross-validation (5×3 CV) [23]. It repeats five times on the complete data (5 outer folds), and each outer fold includes three inner folds, for which the data are split into three equal parts for use, sequentially, used as training, validation, and test sets. The validation sets to tune the hyperparameters for the models, for the vector space, doc2vec and BERT methods. The two first representation models need to be trained on the whole text corpus while

the last model comes with different pre-trained options, so we select the best hyperparameters/pre-train models according to the one with the highest predictive performance on the validation set for each inner fold of the 5×3 CV. For the vector space-based method, the optimal number of l dimensions of truncated SVD are selected according to the scree plot of the profile log-likelihood, or $l = 606$ [44]. For doc2vec, no available method exists to pre-identify the best values for the hyperparameters, so the model needs to be fine-tuned using the 5×3 CV. For BERT, we select the best model based on their performance on the validation set of the 5×3 CV. Table 4 contains the list of text representation hyperparameters.

Finally, we evaluate the performance of the student dropout prediction models, using area under the receiver operating characteristic curve (AUC) and top-decile lift (TDL) metrics. The AUC is a common evaluation metric for binary classification and is popular in student dropout prediction literature [2,8,69]. It reflects the probability that the model correctly ranks a randomly selected dropout student as at higher risk than a randomly selected non-dropout student. Then the TDL, another popular evaluation metric for binary prediction models from the business analytics domain [27,70,71], indicates the relative density of students in the top decile; if it exceeds 1, the density of dropout students in the top decile is higher than the density of dropout students in the whole data set, so the model predicts high risk dropout students better than random guessing.

5. Results and discussion

In presenting our study results, we actively address the research questions. Section 5.1 pertains to the potential benefits of incorporating unstructured, textual data to predict student dropouts, and Section 5.2 outlines the interpretability of the proposed decision support framework, with extensive visualizations and analyses.

Table 4
Hyperparameters for text representation methods.

Text Representation Method	# Models	Hyperparameter	Value Range
vector space	1	• Number of dimensions of SVD (decided by Profile Log-likelihood)	$l \in [1, 4996]$
doc2vec	36	• Vector size • Window size • Minimum word frequency • The training algorithm • Negative sampling	25, 50 1, 2, 5 1 PV-DM, PV-DBOW 10, 15, 20
BERT	3	• Pre-trained BERT model	BERT Base, Small BERT, ALBERT

Table 3
List of models and data combinations.

Group	No.	Model name	Student Group		Data Type		Text Representation Method			Segmentation with Textual Data	
			With text (T)	Without text (NT)	Structured (S)	Unstructured (U)	vector space (VS)	doc2vec (DV)	Transformer language model (BERT)	k-means (KM)	Hierarchical clustering (HC)
1	1	Mod_FD_S	✓	✓	✓						
	2	Mod_T_S	✓		✓						
	3	Mod_NT_S		✓	✓						
	4	Mod_TNT_S	✓	✓	✓						
2	5	Mod_T_U	✓			✓	(a) ✓	(b) ✓	(c) ✓		
3	6	Mod_T_SU	✓		✓	✓	Best method from Group 2				
	7	Mod_T_SU_SEG	✓		✓	✓				(a) ✓	(b) ✓

5.1. RQ1: Does incorporating textual feedback data benefit student dropout predictions?

To answer RQ1, we consider three sub-questions, step by step, and thus investigate the benefits of using the feedback text as a simple indicator to separate student groups (RQ1a), using only textual feedback data as predictors (RQ1b), and integrating textual feedback data with traditional structured data (RQ1c).

5.1.1. RQ1a: Modeling students with and without textual feedback separately

With the models in Group 1, we test for any noticeable difference in dropout prediction between groups of students, based on whether they offer textual data or not. This binary indicator enabled us to create two basic segments, to which we apply four models trained only on structured data. The first, *Mod_FD_S*, is trained with full structured data from both student groups, that is, with and without feedback text. To compare *Mod_FD_S* with other models, we extract predictions based on the test sets of the two different groups of students, with and without feedback text, for which *Mod_FD_S_testT* and *Mod_FD_S_testNT* represent two additional variances for each group of students, respectively. Then the second and third models, *Mod_T_S* and *Mod_NT_S*, are trained independently on the structured data from either student group. Finally, *Mod_TNT_S* is a concatenation of *Mod_T_S* and *Mod_NT_S*, which represents a simple segmentation approach for a comparison with the first model. Table 5 summarizes the predictive performance of these models, according to their AUC and TDL; Table 6 provides the results of Wilcoxon signed-rank tests.

To assess the general effect of modeling student groups with and without textual feedback separately, we compare *Mod_FD_S* with *Mod_TNT_S*. According to Table 5, the average performance of model *Mod_TNT_S* with two student segments (AUC = 0.750, TDL = 3.493) is lower than that of *Mod_FD_S* (AUC = 0.755, TDL = 3.818). The Wilcoxon test in Table 6 instead indicates that these differences are only significant for TDL ($p_{TDL} < 0.1$), not for AUC ($p_{AUC} > 0.1$). When we compare the predictive performance of these two models for each particular group of students, with and without textual feedback, the results in Table 5 further indicate that the models trained with the full structured data, *Mod_FD_S_testT* (AUC = 0.791, TDL = 4.225) and *Mod_FD_S_testNT* (AUC = 0.709, TDL = 3.305), attain slightly higher AUC and TDL than *Mod_T_S* (AUC = 0.784, TDL = 3.947) and *Mod_NT_S* (AUC = 0.704, TDL = 3.049). Only the TDL difference between *Mod_FD_S_testNT* and *Mod_NT_S* is significant though (Table 6). Therefore, the use of a binary signal to segment students, according to whether they offer textual feedback, does not improve predictions for the respective groups.

5.1.2. RQ1b: Information in unstructured, textual data

The preceding findings suggest the need to consider unstructured, textual data. To assess its incorporation, we zoom in on the specific group of students that provide feedback text, T. Table 7 contains the predictive performance results for Group 2, model *Mod_T_U* with three text representation methods corresponding to three sub-models trained only on unstructured, textual data, *Mod_T_U_VS*, *Mod_T_U_DV* and *Mod_T_U_BERT*, which represent the data using the vector space, doc2vec and BERT approaches, respectively. The best method is then selected

Table 5

Results of the dropout prediction models (average and standard deviation of AUC and TDL), Group 1.

Group	No.	Model Name	AUC (SD)	TDL (SD)
1	1	<i>Mod_FD_S</i>	0.755 (0.046)	3.818 (0.872)
	1a	<i>Mod_FD_S_testT</i>	0.791 (0.043)	4.225 (0.974)
	1b	<i>Mod_FD_S_testNT</i>	0.709 (0.065)	3.305 (0.916)
	2	<i>Mod_T_S</i>	0.784 (0.048)	3.947 (1.186)
	3	<i>Mod_NT_S</i>	0.704 (0.064)	3.049 (0.840)
	4	<i>Mod_TNT_S</i>	0.750 (0.044)	3.493 (0.866)

Table 6

Wilcoxon signed-rank test for pairwise comparison of models in Group 1.

	Model Name	Model Name	p-Value	
			AUC	TDL
Group 1 vs. 1	<i>Mod_T_S</i>	vs. <i>Mod_FD_S_testT</i>	0.524	0.100*
	<i>Mod_NT_S</i>	vs. <i>Mod_FD_S_testNT</i>	0.489	0.449
	<i>Mod_TNT_S</i>	vs. <i>Mod_FD_S</i>	0.561	0.055*

Notes: Significant differences at the 90%, 95%, and 99% levels are indicated by *, **, and ***.

Table 7

Results of the dropout prediction models (reported on average and standard deviation of AUC and TDL), Group 2. The best hyper-parameter combination is highlighted.

Group	No.	Model Name	AUC (SD)	TDL (SD)
2	–	<i>Random guessing</i>	0.500 (0.000)	1.000 (0.000)
	5a	<i>Mod_T_U_VS</i>	0.539 (0.046)	1.213 (0.369)
	5b	<i>Mod_T_U_DV</i>	0.591 (0.030)	1.656 (0.625)
	5c	<i>Mod_T_U_BERT</i>	0.545 (0.066)	1.467 (0.512)

for model *Mod_T_U*. As a comparison baseline, we also include a random guess model. Table 8 reports the findings of Wilcoxon signed-rank significant tests.

According to the results in Table 7, *Mod_T_U_VS* (AUC = 0.539, TDL = 1.213), *Mod_T_U_DV* (AUC = 0.591, TDL = 1.656) and *Mod_T_U_BERT* (AUC = 0.545, TDL = 1.467) all perform better than random guessing. Comparing the performance in terms of AUC and TDL of these models, we observe that the doc2vec method performs better than the vector space and BERT methods for predicting students at risk. Table 8 confirms that the differences between model *Mod_T_U* (using doc2vec method) and random guessing are significant, so the unstructured, textual data provide valuable information for predicting dropout students ($p < 0.01$).

Although these results clearly indicate that there is valuable information in unstructured, textual data, they are not sufficient on their own to build a good predictive model; they should be incorporated with structured data. Considering the evidence that the doc2vec text representation method performs better than vector space and BERT, we focus on doc2vec in our continued analyses.

5.1.3. RQ1c: Incorporating unstructured, textual data and its segmentation layer

For this sub-question, we address the most important part of RQ1, namely the benefits of incorporating unstructured, textual data and the segmentation approach. With doc2vec, we build segmented prediction models (for the detailed process, see Section 3.1). Here, we investigate the predictive performance of the models in Group 3, *Mod_T_SU* and *Mod_T_SU_SEG*, where the first setup integrates both structured and unstructured data, whereas the latter represents the final combination of *Mod_T_SU* with an additional layer of segmentation, on top of the student feedback text. Furthermore, we compare two setups of model *Mod_T_SU_SEG* using k-means (*Mod_T_SU_KM*) and hierarchical clustering (*Mod_T_SU_HC*) method to select the best one. The two relevant models from Group 1 provide a benchmark. We present the results in Table 9 and the pairwise statistical tests for the models in Table 10.

Table 8

Wilcoxon signed-rank test for pairwise comparisons of models in Group 2.

	Model Name	Model Name	p-Value	
			AUC	TDL
Group 2 vs. 2	<i>Mod_T_U</i>	vs. <i>Random guessing</i>	<0.001***	0.003**

Notes: Significant differences at the 90%, 95%, and 99% levels are indicated by *, **, and ***.

Table 9

Results of the dropout predictive models (reported on average and standard deviation of AUC and TDL), Group 3. The best hyper-parameter combination is highlighted.

Group	No.	Model Name	AUC (SD)	TDL (SD)
1	1a	<i>Mod_FD_S_testT</i>	0.791 (0.043)	4.225 (0.974)
	2	<i>Mod_T_S</i>	0.784 (0.048)	3.947 (1.186)
	6	<i>Mod_T_SU</i>	0.784 (0.045)	4.181 (0.799)
3	7a	<i>Mod_T_SU_KM</i>	0.801 (0.035)	4.524 (0.872)
	7b	<i>Mod_T_SU_HC</i>	0.788 (0.082)	4.188 (1.224)

Notes: Group 1 provides a benchmark.

Table 10

Wilcoxon signed-rank test for pairwise comparison of models in Groups 1 and 3.

		Model Name	Model Name	p-Value	
				AUC	TDL
Group 3 vs. 3	<i>Mod_T,SU</i>	vs.	<i>Mod_T,SU_SEG</i>	0.018**	0.038**
Group 3 vs. 1	<i>Mod_T,SU_SEG</i>	vs.	<i>Mod_T,S</i>	0.022**	0.062*
	<i>Mod_T,SU_SEG</i>	vs.	<i>Mod_FD,S_testT</i>	0.208	0.087*

Notes: Significant differences at the 90%, 95%, and 99% levels are indicated by *, **, and ***.

To find the best setup for incorporating unstructured, textual data with structured data, we compare *Mod_T_SU* with its segmented modeling versions *Mod_T_SU_KM* and *Mod_T_SU_HC*. As Table 9 shows, the average performance of both *Mod_T_SU_KM* (AUC = 0.801, TDL = 4.524) and *Mod_T_SU_HC* (AUC = 0.788, TDL = 4.188) is better than that of *Mod_T_SU* (AUC = 0.784, TDL = 4.181). Additionally, we find that the model using k-means segmentation method perform better than hierarchical clustering in terms of AUC and TDL. Therefore it is selected as the final method for *Mod_T_SU_SEG*. Table 10 confirms that the 2.2% increase in AUC and 8.2% improvement in TDL are both significant ($p < 0.05$). Thus, the unstructured, textual data provide a good segmentation base for improving the performance of the student dropout prediction model.

When we compare the performance of the best model in Group 3 with the models trained only on structured data in Group 1, we specify the benefits of incorporating unstructured, textual data. According to Tables 9 and 10, *Mod_T_SU_SEG* using k-means method (AUC = 0.801, TDL = 4.524) outperforms the baseline model *Mod_T_S* (AUC = 0.784, TDL = 3.947) for both AUC ($p = 0.022$) and TDL ($p = 0.062$). Moreover, *Mod_T_SU_SEG*, trained only on the specific group of students with feedback text (about half of them), performs even better than *Mod_FD_S_testT* (AUC = 0.791, TDL = 4.225), which is trained on all students. The difference is significant for TDL ($p_{TDL} < 0.1$).

In summary, three sub-questions, RQ1a–1c, together inform RQ1. As we find by testing RQ1a, simply using the information that a student gives feedback, as a binary indicator, is not enough to improve student dropout predictions, and more advanced incorporation techniques are needed. With RQ1b and RQ1c, we confirm that there is valuable information in the unstructured, textual feedback data, which can create meaningful student segments and significantly improve the predictive performance of the dropout application. In conclusion, the student dropout prediction application works better for students who provide feedback text, and institutions and academic decision-makers should collect and incorporate more unstructured, textual data from students.

5.2. RQ2: What insights can be derived from the incorporated textual data to help decision-makers better understand and manage student dropout?

To help decision-makers understand different dropout drivers per student group and develop more personalized different retention strategies, we focus the best modeling model approach, *Mod_T_SU_SEG*,

which uses a doc2vec text representation method and k-means segmentation algorithm. The results of the model come from the first fold of the $5 \times 3CV$. Then, to evaluate and interpret the incorporated textual data for each student, we use four visualization tools and expert labeling. First, we construct a visualization based on the vectors of the doc2vec transformed text to evaluate the separation of the student segments [54]. Second, to extract segment-level insights, we set up an expert labeling task, in which three independent experts assign the top bigram words of each student segment to four categories that reflect different feedback dimensions: student, teacher, course, and facility. The final assignment of each bigram reflects a majority vote and agreement among the three experts, which we confirm with an interrater reliability (IRR) analysis based on Fleiss's Kappa index [72,73]. According to the IRR, the experts achieve strong agreement, with $kappa > 0.9$, $p < 0.001$, for both student segments.

For the selected CV fold, the k-means algorithm and doc2vec methods create two segments of students. Fig. 4 shows the transformed feedback text for each student segment, reduced to two dimensions using PCA, t-SNE, t-SNE with PCA as input, and UMAP techniques. The low dimension version appears together with the student segment labels, according to these two-dimensional scatter plots. This visualization clearly reveals that the two student groups can be well segmented, with very little overlap. Therefore, the segmentation algorithm in the proposed decision support framework successfully detects two meaningful student groups in the data.

Fig. 5 presents the profiles of the two student segments. The top 100 bigrams of each segment are extracted and labeled, according to student, teacher, course, and facility categories. Then for each student segment, we calculate the percentage difference in bigrams per category, relative to the overall percentage of two segments in the same category. Thus, we determine which topics students describe relatively more in each segment, compared with all students on average. According to Fig. 5, students in segment 1 care more about teachers (2.43% above average), student life (0.65% above average), and school facilities (0.56% above average) but less about courses (3.65% below average). The interests of the students in segment 2 reflect the opposite trends, such that they talk more about the courses (3.65% above average) but less about the teachers, student life, or school facilities (2.43%, 0.65% and 0.56% below average, respectively).

As a further assessment of student segments, we analyze the variables in the LR models to derive the importance of each variable group per student segment (see Fig. 6). For this calculation, we multiple the coefficient of the LR model and the standard deviation of the corresponding input variable, then take the absolute value. Summing these importance values of the group and dividing by total importance provides the percentage of variable importance per group. According to Fig. 6, students in segment 1 exhibit higher dropout percentages (1.9%) than those in the second segment (0.8%). In student segment 1, information related to course enrollment (46%) and feedback text (17%) contribute more to dropout predictions than this information does for student segment 2 (43% and 14%, respectively). Sociodemographic information indicates opposite effects.

Finally, in Fig. 7 we present the coefficients of the LR model for structured data for each segment of students. Using the findings we have gathered thus far, we can derive a full picture of dropped out students and strategies for communicating with them. The students at risk in segment 1 mostly study on the Paris campus; have changed their address at least once; enrolled in the course categories ACC, ECO, LAN, LAW, or QMS in their first semester (see Fig. 7 for the full names of each category); and provide feedback mostly about teachers, student life, and facilities. Students at risk in segment 2 instead are mostly international students; enrolled in the course categories ACC, MIS, QMS, STR, or RES in the first semester; and share a tendency to offer feedback about courses. Therefore, the HEI should identify students with these profiles and learn what difficulties they face, on the basis of their feedback, before initiating an intervention.

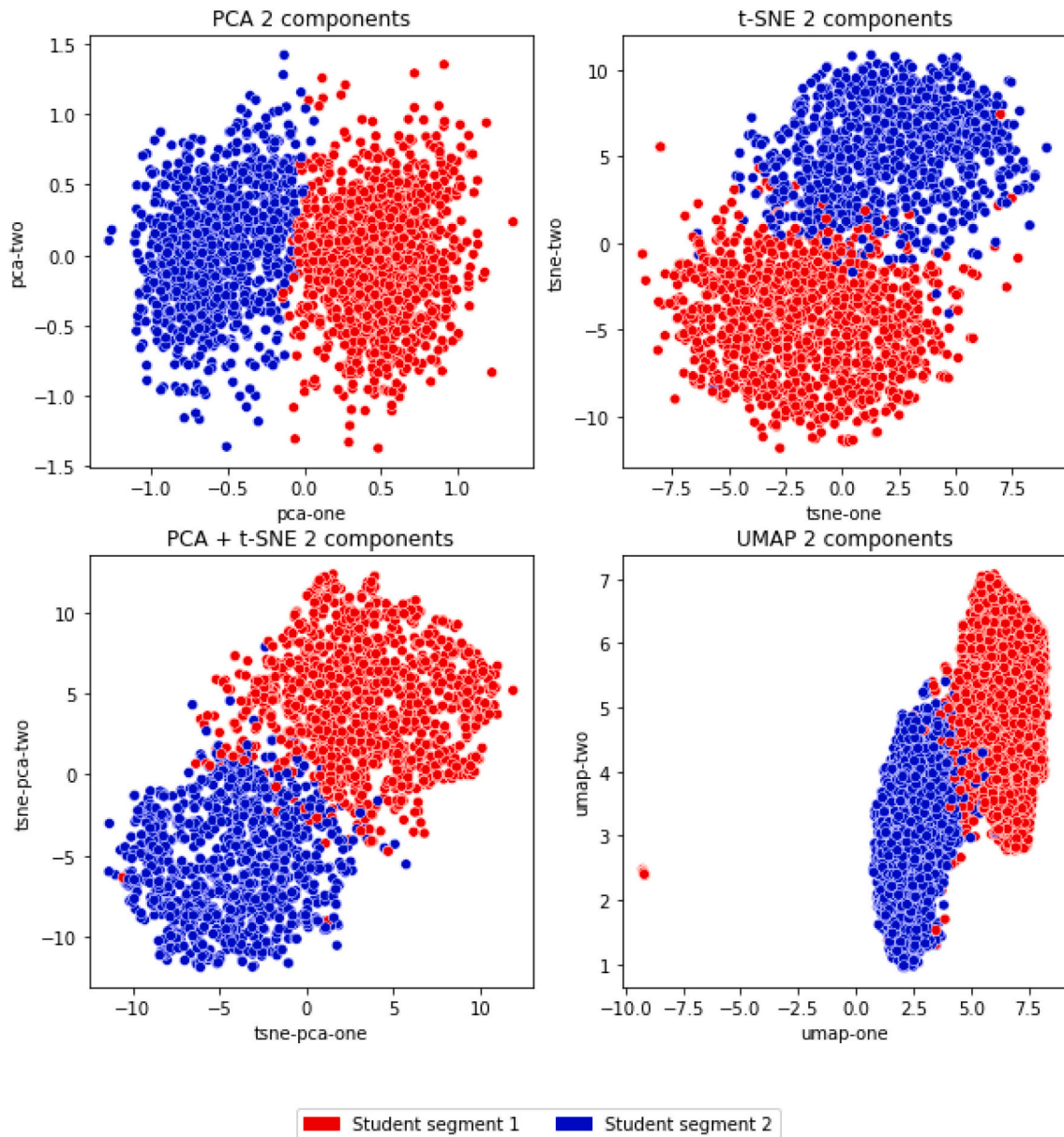


Fig. 4. Visualization of the student segments based on transformed textual data.

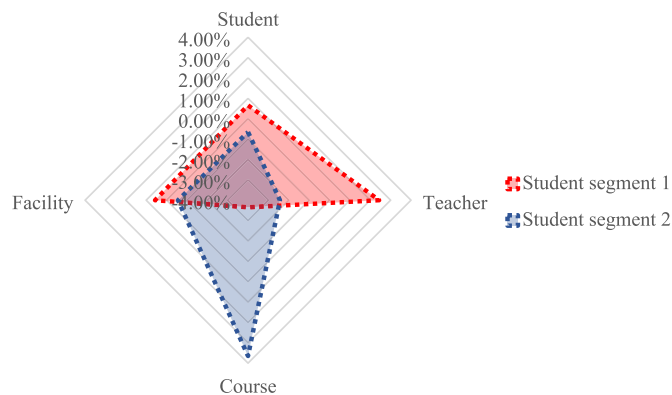


Fig. 5. Discussion topics among the two student segments.

6. Conclusions

This study proposes a new decision support framework to incorporate both structured and unstructured, textual data and thereby improve student dropout applications, in terms of both their predictive performance and their interpretation. Applying this framework to a real-life data set, obtained from a French HEI, reveals that textual feedback data contain valuable information that can be used to detect at-risk students. By integrating these data, we significantly improve the predictive performance of the student dropout prediction model. Moreover, unstructured, textual data can be leveraged to create meaningful student segments. The proposed approach outperforms existing methods, in terms of predictive performance, and successfully identifies student segments that require distinct predictive models. From an interpretability perspective, because it can analyze student segments, our proposed method offers informative insights about student concerns and communication habits. These findings can be used to design and adapt interactions with different groups of students.

This study extends research that identifies segmented modeling

Segment Definition			Variable Category Importance		
Seg.	Number of observations	Percentage dropout	Sociodemographic information	Course enrollment	Feedback text
1	2,182	1.9%	37%	46%	17%
2	2,635	0.8%	43%	43%	14%

Fig. 6. Percentage of variable importance per group in each segment.

Seg.	Variable Group																								
	Sociodemographic Information								Course Enrollment																
	Age	Origin: Others	Region: 92	Region: Others	Civil status: B	Civil status: Others	Campus: Paris	Info. changed	Total	ACC	DEV	ECO	ENT	FAS	FIN	IBE	LAN	MIS	LAW	MKT	NEG	QMS	STR	RES	OPS
1	-								-	+		+			-		+	-	+		-	+			-
	0.33								0.05	0.02		0.59			0.51		0.35	0.23	0.08		0.54	0.72			0.40
2	-	+	-	-	-		-	-	-	+	-		-	-	-	-		+		-	-	+	+	+	
	0.21	0.48	0.43	0.37	0.29		0.11	2.74	0.04	0.04	0.09		0.02	0.14	0.16	0.37		0.13		0.14	0.17	1.22	0.07	0.28	

Fig. 7. Coefficients of the variables in each student segment.

Notes: The course category abbreviations are as follows: audit, control, accounting (ACC), personal development (DEV), economics (ECO), entrepreneurship (ENT), fashion (FAS), finance (FIN), international economy (IBE), languages (LAN), management of information systems (MIS), law (LAW), marketing (MKT), negotiation (NEG), quantitative methods (QMS), strategy (STR), research (RES), operation management (OPS).

approaches to improve predictive performance and interpretability in predictive modeling tasks [3,27]. By adding another layer of student segments, we significantly improve the predictive performance of the model. Building segmentation on unstructured, textual information reveals informative segments, which provide clearer understanding of heterogeneity in the data set. In turn, our segmentation approach suggests avenues for further research related to decision support system and learning analytics. First, our approach might be applied to detect student segments using other textual data, such as student submission files, communications on a course page, or student questions shared by email. Second, we test our framework with a unique, real-world data set, but additional tests with different sets of data would help validate its general performance in other settings. Third, the choice of components to include in the framework is flexible. We rely on a set of proven and SOTA methods, but future research can also test and compare between other text representation, segmentation, and predictive modeling methods.

CRedit authorship contribution statement

Minh Phan: Conceptualization, Data curation, Formal analysis, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. **Arno De Caigny:** Conceptualization, Data curation, Formal analysis, Methodology, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. **Kristof Coussement:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

The authors declare no conflict of interest.

Data availability

The data that has been used is confidential.

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